**University of Essex**
**MSc Artificial Intelligence**
**Module: KRAR_PCOM7E October 2025**
**Unit 11: Collaborative Discussion – Ontology Language Comparison**

## Initial Discussion

In *Lecture notes in business information processing*, Kalibatiene and Vasilecas assert that “[a]n ontology is a formal, explicit specification of a shared conceptualization” (Kalibatiene, Vasilecas, 2011). The question at hand is to evaluate which of IF, OWL 2, OWL Lite, or RDF is the most useful development platform for software agents running on the Internet.

Of the four options, my view is that OWL 2 is the most appropriate for intelligent agents that are required to operate online or interact with other online intelligent agents. The reasons are straightforward:

1. As a baseline practical matter, OWL 2 is defined by a W3 specification, meaning, its implementation details have been formally structured in accordance with software development best practices (W3.org, 2012).
2. Unlike the other options, OWL 2 offers a variety of robust features that allow for quicker and more accurate development of formally-defined class hierarchies, such as property chains, cardinality restrictions, complex class constructors, disjointness syntax, and granular property characteristics such as symmetry and transitivity (W3.org, 2012).
3. Most importantly, and aligned with Kalibatiene and Vasilecas’ definition of an ontology, OWL 2 requires formal standardization of class characteristics as a prerequisite for using them for logical reasoning, which reduces the chance of errors; the other options (RDF in particular) are more permissive, which may increase the chance of introducing logical reasoning errors (Iglesias-Molina, 2022; Kalibatiene, Vasilecas, 2011; W3.org, 2012).

Finally, it should be noted that this determination should not be taken to mean that OWL 2 will always be the best choice for any given use case; the choice of which ontological development platform to use must rest on the specific fact pattern underlying the use case at hand (Iglesias-Molina, 2022).

## References

Iglesias-Molina, A., Cimmino, A., Ruckhaus, E., Chaves-Fraga, D., Raúl García-Castro and Corcho, O. (2022). An ontological approach for representing declarative mapping languages. *Semantic Web*, 15(1), pp.191–221. doi:https://doi.org/10.3233/sw-223224.

Kalibatiene, D. and Vasilecas, O. (2011). Survey on Ontology Languages. *Lecture notes in business information processing*, pp.124–141. doi:https://doi.org/10.1007/978-3-642-24511-4_10.

W3.org. (2012). *OWL 2 Web Ontology Language Document Overview (Second Edition)*. [online] Available at: https://www.w3.org/TR/owl2-overview.

## Peer Reponse No. 1

Your post presents a clear and well-structured argument in favour of OWL 2, and I agree that its strong formal foundations make it a compelling choice for intelligent software agents operating on the web. In particular, your emphasis on OWL 2's grounding in W3C standards and its rich support for class hierarchies, property constraints, and logical reasoning aligns well with Kalibatiene and Vasilecas' (2011) definition of ontology as a formal and explicit shared conceptualisation.

I also found your point about OWL 2's stricter requirements for formalisation particularly convincing. Compared to RDF, the enforced semantics and reasoning prerequisites in OWL 2 do indeed reduce ambiguity and improve consistency, which is critical when agents rely on shared knowledge structures for automated inference.

That said, one possible limitation worth considering is the computational and implementation complexity of OWL 2 in large-scale or performance-sensitive web environments. While its expressiveness is advantageous, not all agent-based systems require advanced reasoning features such as property chains or complex class constructors. In such cases, lighter ontology languages may offer better scalability and ease of adoption.

Overall, your conclusion appropriately recognises that no single language is universally optimal. Your post demonstrates strong critical awareness by situating OWL 2 as the most suitable option given specific use-case requirements, rather than as an absolute solution. This balanced perspective strengthens your argument and reflects sound ontological reasoning.

### References

None provided.

## Peer Reponse No. 2

You present a compelling case for OWL 2, and I largely agree with your well-structured reasoning. Your emphasis on its formal W3C standardisation, robust feature set, and requirement for explicit specification is highly persuasive and directly aligns with the demands of the Kalibatiene and Vasilecas (2011) definition. Your third point, regarding error reduction through formalisation versus RDF's permissiveness, is particularly critical for reliable agent interoperability.

To engage constructively with your argument, I would propose considering the practical implications of your final, nuanced point: that the ‘best choice’ depends on the ‘specific fact pattern’. While OWL 2's expressivity is its greatest strength for complex reasoning, this same characteristic can be its primary drawback in specific web agent scenarios. The computational complexity of reasoning with full OWL 2 can be prohibitive for agents operating in real-time or with limited resources (Horridge et al., 2014). In such high-performance or large-scale distributed environments, common on the WWW, the simpler semantics of OWL Lite or even RDF Schema might offer a more useful and efficient foundation for a shared conceptualisation, even if they are less expressive.

Therefore, your argument might be strengthened by briefly acknowledging this trade-off. While OWL 2 is indeed the most *capable* and *rigorous* language for fulfilling the definition in a general sense, its practical *usefulness* for a given agent population is contingent on their reasoning requirements and operational constraints. This does not weaken your conclusion but grounds it in the pragmatic reality of agent-based system design.

### References


Horridge, M., Bail, S., Parsia, B. and Sattler, U. (2011). The Cognitive Complexity of OWL Justifications. *Lecture Notes in Computer Science*, pp.241–256. doi:https://doi.org/10.1007/978-3-642-25073-6_16.

Kalibatiene, D. and Vasilecas, O. (2011). Survey on Ontology Languages. *Lecture notes in business information processing*, pp.124–141. doi:https://doi.org/10.1007/978-3-642-24511-4_10.

## Peer Reponse No. 3

Your argument for OWL 2 is well-founded, particularly regarding its W3C standardisation and formal semantics that reduce reasoning errors. Your acknowledgment that context matters demonstrates appropriate nuance.

Fatema and Deane both raise the issue of computational complexity, which deserves emphasis. As Horrocks et al. (2003) note, OWL DL reasoning is NEXPTIME-complete, making it computationally expensive for resource-constrained or real-time scenarios. Dentler et al. (2011) empirically demonstrate that even moderately-sized OWL 2 ontologies can require substantial reasoning times, potentially rendering them impractical for certain web applications despite their expressiveness.

However, I would suggest that OWL 2 profiles (EL, QL, RL) partially address these scalability concerns while retaining significant expressiveness beyond OWL Lite or RDFS (W3C, 2012). OWL 2 EL, for instance, supports polynomial-time reasoning while accommodating property chains and complex class constructors, features Matthew rightly identifies as valuable for accurate ontology development.

Deane's point about the "specific fact pattern" is critical: the optimal choice depends on whether an application prioritises expressiveness for complex reasoning or computational efficiency for scalability. As Krötzsch et al. (2012) argue, this trade-off between logical expressiveness and reasoning tractability fundamentally shapes ontology language selection.

This discussion effectively demonstrates that while OWL 2 offers superior formal capabilities, practical utility must account for computational constraints and operational requirements in distributed web environments.

### References

Dentler, K., Cornet, R., ten Teije, A. and de Keizer, N. (2011). Comparison of reasoners for large ontologies in the OWL 2 EL profile. *Semantic Web*, 2(2), pp.71–87. doi:https://doi.org/10.3233/sw-2011-0034.

Horrocks, I., Patel-Schneider, P. F. and van Harmelen, F. (2003). From SHIQ and RDF to OWL: the making of a Web Ontology Language. *Journal of Web Semantics*, 1(1), pp.7–26. doi:https://doi.org/10.1016/j.websem.2003.07.001.

Krötzsch, M., Simancik, F. and Horrocks, I. (2026). *A Description Logic Primer*. [online] arXiv.org. Available at: https://arxiv.org/abs/1201.4089 [Accessed 19 Jan. 2026].

## Summary Post

In my initial discussion post, I asserted that the OWL 2 ontology development platform is currently the most appropriate default choice for Internet-enabled automated agents, especially when viewed in context of Kalibatiene and Vasilecas' definition of an ontology as a "formal, explicit specification of a shared conceptualization" (Kalibatiene, Vasilecas, 2011).

I offered three reasons to support my assertion:

1. The OWL 2 technical specification was created by the W3 in accordance with industry best practices, making it more readily amenable to integration with other industry-standard platforms in may need to interact with (W3.org, 2012).
2. OWL 2 offers a variety of features not available in other platforms that allow for more efficient and productive development of ontological class hierarchies, such as polynomial-time reasoning over property hierarchies and class constructors (W3.org, 2012).
3. OWL 2 is less likely to result in implementation and runtime errors, since it requires formal standardization of class characteristics (Iglesias-Molina, 2022; Kalibatiene, Vasilecas, 2011; W3.org, 2012).

These three reasons were offered with the caveat that the specific end use goals of an ontology should determine the choice of its development platform more than any other factor (Iglesias-Molina et al, 2022).

Peer responses largely agreed with this assessment, while also raising the highly salient point that OWL 2 may encounter the practical drawbacks of increased computational resource consumption and/or implementation complexity when used at scale or in latency-sensitive runtimes (Dentler et al, 2011; Horridge et al, 2011; Horrocks, Patel-Schneider, and van Harmelen, 2003).

In such use cases, or when faced with such practical implementation constraints, peer response then asserted that OWL Lite and/or RDF Schema could be more resource-efficient platforms, despite the trade-off of reduced expressiveness of ontological class structures and relationships in comparison to OWL 2 (Horridge et al, 2011; Krötzsch, Simancik and Horrocks, 2026).

Overall, I am in agreement with the peer responses that any choice of an ontology development platform should be made with such practical considerations in the forefront of the software selection process.

## References


Dentler, K., Cornet, R., ten Teije, A. and de Keizer, N. (2011). Comparison of reasoners for large ontologies in the OWL 2 EL profile. *Semantic Web*, 2(2), pp.71–87. doi:https://doi.org/10.3233/sw-2011-0034.

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Iglesias-Molina, A., Cimmino, A., Ruckhaus, E., Chaves-Fraga, D., Raúl García-Castro and Corcho, O. (2022). An ontological approach for representing declarative mapping languages. *Semantic Web*, 15(1), pp.191–221. doi:https://doi.org/10.3233/sw-223224.

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